

Chapter 106 — Evaluation and Management of Males With Urinary Incontinence — Numbers, Gold Standards and Traps

IF YOU READ NOTHING ELSE

1. **AUS (artificial urinary sphincter)** = **gold standard**, moderate-severe SUI; $\approx 80\%$ continent.
2. **Wait ≥ 6 –12 months** after radical prostatectomy before SUI surgery.
3. **PFMT (pelvic floor muscle training)** ≥ 3 months, supervised $\rightarrow$ first line SUI.
4. UUI conservative: **≥ 6 weeks, ideally 3 months**; cut fluid 25% if >1 L/day.
5. **Antimuscarinics + β_3 -agonists** = first-line drugs for UUI.
6. **UDS (urodynamics)** **only if it changes management** — before invasive therapy.
7. **OnabotulinumtoxinA 100 U** idiopathic OAB; **200 U** neurogenic DO; retention $\approx 10\%$.
8. **SNM (sacral neuromodulation)**: S3; proceed if **$\geq 50\%$ improvement**.
9. **Bulking agents: do not offer** for post-prostatectomy incontinence.
10. **MASTER**: sling = AUS at 12 months; AUS better on secondary outcomes.

URINARY INCONTINENCE

- **UI (urinary incontinence)**: ICS (International Continence Society) — **any involuntary loss of urine**; a storage symptom.
- **Most important continence determinant**: **EUS** (external urethral sphincter).
- **Major independent risk factor**: **age**.
- **DIAPPERS**: Delirium, Infection, Atrophic urethritis, Pharmacological, Psychiatric, Endocrine, Reduced mobility, Stool impaction.
- **Male:female** $\approx 1:2$; prevalence 1–40%; 11–34% in older men.
- **Commonest male type**: **UUI** (urgency) $>$ SUI (stress) $>$ MUI (mixed).
- **EpiLUTS 46%** — inflated by broad definition.

TYPES OF URINARY INCONTINENCE

- **UUI $\rightarrow$ DO (detrusor overactivity)** on UDS in $\approx 90\%$ of men.
- **Male SUI $\rightarrow$ almost always post-prostatectomy**.
- **Adult new nocturnal enuresis $\rightarrow$ exclude high-pressure chronic retention**.
- **Enuresis** = wet asleep, unaware; **nocturia** = wakes to void.
- **Significant residual**: ≥ 300 mL.
- **Chronic retention + SCI (spinal cord injury)** $\geq T6 \rightarrow$ **autonomic dysreflexia**.
- **Continuous UI $\rightarrow$ fistula, exstrophy/epispadias, ectopic ureter**.
- **Enuresis prevalence**: $\approx 10\%$ at age 7; persists in 2–3%.

EVALUATION

- **Essential**: history, exam, BMI, dipstick, diary, questionnaire, flow, PVR (postvoid residual).
- **Optional**: PSA, bloods, imaging, endoscopy, invasive UDS.
- **Cough test**: supine, then **standing**.
- **AUS balloon 61–70 cm H_2O** ; abdominal pressure above $\rightarrow$ leak.
- **Only fully validated diary**: **ICIQ-BD** (ICI Questionnaire Bladder Diary).
- **Diary length**: **3 days = 7 days**; include weekend.
- **ICIQ-UI-SF** (short form): score items 3+4+5 = **0–21**; separates SUI vs UUI.
- **IPSS** (International Prostate Symptom Score) **does not measure incontinence**.
- **Pad count**: 1 mild; 2–4 moderate; >4 severe.
- **Pad weight/day**: <200 g mild; 200–400 g moderate; >400 g severe — better measure; unvalidated.
- **Pad test** not recommended by most guidelines; **cannot separate UUI/SUI**.

- **Flow**: fast rise $\rightarrow$ OAB (overactive bladder); slow rise, long tail $\rightarrow$ BOO (outflow obstruction).
- **Flow cannot separate BOO vs detrusor underactivity**.
- **Flows**: ≥ 2 traditionally; **1 if representative** (ICS 2017).
- **PSA**: if life expectancy >10 years and result changes management.
- **Bloods**: high PVR or diabetes $\rightarrow$ renal function, glucose.
- **Cystourethroscopy if**: stricture, BOO, haematuria, high PVR.
- **Imaging only if**: upper-tract concern, high PVR, stones.
- **UDS gold standard** for LUT (lower urinary tract) function.
- **Stress test in UDS**: Valsalva/cough **upright at 200 mL**, repeat at capacity.
- **USI (urodynamic stress incontinence)** vs cough-associated DOI $\rightarrow$ **refill supine, cough**.

MANAGEMENT — URGENCY INCONTINENCE

- **Antimuscarinics**: 7 agents; all **level 1, grade A**; similar efficacy; **ER $>$ IR**.
- **Old rule**: ≥ 2 agents, ≥ 4 weeks each; now one then move on.
- **Solifenacin $>$ tolterodine**: efficacy, less dry mouth.
- **Trospium**: quaternary amine $\rightarrow$ **no cognitive effects**.
- **Anticholinergic ≥ 3 months** $\rightarrow$ dementia risk $\uparrow \approx 46\%$.
- **Cognitive impairment $\rightarrow \beta_3$ -agonist first** or early surgery.
- **Mirabegron**: monitor **blood pressure**; superior to tolterodine.
- **Vibegron 75 mg daily**; **no CYP interaction**, no blood-brain barrier crossing.
- **Combination**: mirabegron 50 + solifenacin 5 $>$ solifenacin escalation.
- **Tadalafil**: only FDA-approved PDE5I, voiding LUTS + erectile dysfunction.
- **UDS before toxin/SNM**: not mandatory; **FUTURE** trial results awaited.
- **Botox repeat every 6–9 months**; must accept self-catheterisation.
- **Abobotulinum (Dysport) 2.5:1** conversion; more retention.
- **SNM test**: temporary lead 1–2 weeks; tined lead 2–4 weeks.
- **SNM battery**: old 5–7 years; new ≥ 15 years; **Axonics** first MRI-safe rechargeable.
- **SNM** = most effective, durable, **most expensive**.
- **PTNS** (tibial): behind **medial malleolus, S2–S4**; **30 min weekly $\times 12$** .
- **PTNS** = least invasive; better for **dry OAB**.
- **Last resort**: augmentation enterocystoplasty or ileal conduit.
- **Autoaugmentation**: no longer recommended.

MANAGEMENT — STRESS INCONTINENCE

- **Duloxetine:** off-label in men; Weak offer, Strong inform.
- **Before surgery:** exclude stricture, bladder dysfunction.
- **AUS first described:** Scott, 1973.
- **Mild-moderate:** fixed transobturator sling (Weak).
- **Moderate-severe: AUS (Strong).**
- **AUS/ProACT** → expert centres only.
- **Sling outcome worse:** severe UI, pelvic radiotherapy, prior TURP (transurethral surgery).
- **ProACT contraindicated** after pelvic radiotherapy.
- **AdVance:** transobturator, under membranous urethra.
- **ATOMS:** silicone cushion, noncircumferential; 2–4 adjustments; > **ProACT**.
- **Fixed vs adjustable sling:** neither superior.
- **MASTER:** only 15% never leaked after either operation.
- **Interim containment:** avoid long-term urethral catheter (traumatic hypospadias).

MIXED AND OTHER TYPES

- **MUI:** treat most bothersome first; treat DO → repeat UDS.
- **Retention + enuresis:** catheterise first, then treat BOO.
- **Postmicturition dribble:** no drug; **urethral milking** + pelvic squeeze.
- **Milking:** 3 fingerbreadths behind scrotum; repeat twice.

ALGORITHMS

- **Specialist referral:** haematuria, pain, recurrent UTI, pelvic radiotherapy, abnormal DRE, voiding dysfunction.
- **ICI:** AUS grade A (AMS 800 only); sling C.
- **EAU 2020 algorithm:** pad test Weak; toxin or SNM Strong.
- **Underactive detrusor + UUI** → intermittent catheterisation + drug.

CLASSIC TRAPS

- **Trap:** "overflow incontinence" — **Actually:** incontinence associated with chronic retention.
- **Trap:** SUI = USI — **Actually:** SUI symptom; USI urodynamic finding.
- **Trap:** cough leak = SUI — **Actually:** may be cough-associated DOI.
- **Trap:** terminal dribble postmicturition — **Actually:** voiding symptom.
- **Trap:** IPSS grades incontinence — **Actually:** it does not.
- **Trap:** pad count grades severity — **Actually:** pad weight better.
- **Trap:** UDS before everything — **Actually:** only if it changes management.
- **Trap:** early SUI surgery post-RP — **Actually:** wait 6–12 months.
- **Trap:** bulking for PPI — **Actually:** do not offer.
- **Trap:** ProACT after radiotherapy — **Actually:** contraindicated.